Yousuf Abdullah Borna

Software Engineer | Generative AI | Backend Systems

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Professional Summary

Software Engineer and M.S. Computer Science graduate with enterprise development and AI/ML research experience building RAG systems, agentic workflows, backend APIs, and AI-powered applications. Experienced across Java/Spring Boot and Python-based systems, with hands-on work in retrieval pipelines, embeddings, LangGraph workflows, REST integrations, testing, debugging, and production support.

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, SQL
AI / GenAI: LangChain, LangGraph, RAG, LLM APIs, Agentic AI, FAISS, Embeddings, Semantic Search, Structured Outputs
Backend: Spring Boot, REST APIs, JPA/Hibernate, FastAPI, Node.js, Express.js
Databases: PostgreSQL, Redis, MongoDB, MySQL, SQLite
Cloud / DevOps: AWS Lambda, Docker, GitHub Actions, Git, Maven, Gradle, Linux

AI & Research Experience

Graduate Research Assistant Aug 2024 – May 2026
AI/ML & Software Engineering Research – California State University, Fullerton *Fullerton, CA*

- Built a Python RAG pipeline to reduce weakly grounded mentor responses by chunking mentor content, generating embeddings, retrieving relevant passages, and grounding LLM prompts in source material.
- Developed a vector-similarity recommendation workflow to improve mentor discovery by combining student profile attributes with semantic similarity, enabling more personalized ranking of mentors and resources based on interests and goals.
- Evaluated text preprocessing, embeddings, retrieval logic, and multiple prompt/context combinations to improve the reliability of downstream AI features and identify stronger grounding strategies.
- Integrated AI-enabled functionality with React/TypeScript applications and backend APIs, including structured event logging and AWS Lambda workflows, enabling the research team to capture user interactions and analyze engagement with recommendations and conversational features.

AI Engineering Projects

AgentGuard – AI Agent Security, Governance & Observability Platform **GitHub**
Java, Spring Boot, Python, LangGraph, PostgreSQL, Redis, React, Docker

- Developing AgentGuard to address security and reliability gaps in autonomous AI-agent systems, where agents can invoke tools, access sensitive resources, and operate without sufficient policy enforcement or execution visibility.
- Building authentication, RBAC, policy enforcement, audit logging, and rate limiting to reduce unauthorized agent actions and provide traceability across sensitive agent and tool operations.
- Designing Redis-backed caching and rate limiting, resilience patterns, structured logging, metrics, and distributed tracing across Spring Boot and Python/LangGraph services to improve detection, diagnosis, and isolation of latency issues, failures, and abnormal agent behavior.

Autonomous Clinical Documentation Assistant **GitHub**
Python, LangChain, LangGraph, LLM APIs, FAISS, Pydantic, Streamlit, SQLite

- Built an agentic AI system to reduce the manual effort required to convert clinician-patient conversations into structured documentation, transforming raw conversation input into draft clinical notes and SOAP-style reports for professional review.
- Orchestrated a **4-stage LangGraph workflow** for extraction, summarization, validation, and report generation using structured Pydantic outputs, reducing dependence on a single monolithic prompt and making each processing stage easier to validate and control.
- Implemented FAISS-based RAG and semantic retrieval to reduce reliance on the model's general knowledge, grounding document Q&A in generated reports and stored clinical information.
- Added human-in-the-loop approval so AI-generated documentation is not treated as final until reviewed by a healthcare professional, supporting safer handling of potentially incomplete or incorrect generated content.

Professional Experience

Salesforce Consultant / Developer Jan 2023 – Jun 2024
Eagle Creek / Eviden *Vermillion, SD*

- Developed enterprise applications and REST integrations using Apex, Lightning Web Components, SOQL, Salesforce Flows, and asynchronous processing.
- Automated business workflows and data-processing operations, reducing repetitive manual data-entry activities by approximately **20%**.
- Supported performance optimization, testing, debugging, deployment, and production issue resolution with developers, architects, QA teams, and business stakeholders.

Junior Salesforce Developer Dec 2021 – Dec 2022
Eagle Creek *Vermillion, SD*

- Developed Salesforce functionality using Apex, SOQL, Lightning components, REST APIs, validation rules, and workflow automation while supporting testing, releases, maintenance, and production issue resolution in Agile environments.

Education & Certifications

M.S. in Computer Science, California State University, Fullerton May 2026

B.S. in Computer Science, Siena College 2021

Certifications: Databricks Generative AI Fundamentals | Databricks AI Agent Fundamentals | Salesforce Certified Platform App Builder